SCHOOL OF ENGINEERING,CUSAT

B. TECH DEGREE 1ST SEMESTER SECOND INTERNAL EXAMINATIONS, DECEMBER 2023

23-200-0105B COMPUTER PROGRAMMING(EC A)

(2023 Scheme)

TIME: 2 hrs **MAX. MARKS: 30**

Course Outcomes

On completion of this course the student will be able to:

CO1	Identify main components of a computer system and explain its working
CO2	Develop flow chart and algorithms for computational problems .
CO3	Write the syntax of various constructs of C language.
CO4	Build efficient programs by choosing appropriate decision-making statements,loops and data structures
CO5	Design modular programs using functions for larger problems.

PART A

(Answer all questions)

(5 x 2= 10 Marks)

Q.No.	Questions	Marks	BL	CO	PO
I (a)	Compare linear search and binary search.	2	L2	5	2
(b)	Difference between actual parameter and formal parameter with example.	2	L1	5	1,2
(c)	Differentiate a structure and a union.	2	L1	3	1
(d)	With example briefly explain any 3 string handling functions	2	L3	5	1
(e)	define function and recursive function .	2	L1	5	1

PART B

(Answer the following questions choosing either the first OR second option)

					2x 10= 20 Marks)
Q.No.	Questions	Marks	BL	CO	PO
II (a)	Write a c program to compare two strings without using string functions	5	L3	CO3	3
(b)	Write a c program to print the factorial of a number using recursion.	5	L5	CO5	3,5
	Or				

III (a)	Write a c program to find the smallest element in an array.	5	L3	CO4	3,5
(b)	Write a program to print the marklist of n students using an array of structures. Your program should read and store register number, name and marks for 3 subjects for each student. Display the marklist containing Register number, Name and overall grade in a pleasing format (90-100 -> A grade, 80-89 -> B grade, below 80 -> C grade).	5	L3	CO4	3,5
IV (a)	Illustrate the difference between call by value and call by reference with suitable examples for each.	6	L4	CO5	1,3
(b)	Write a program to check if two strings are equal or not.	4	L3	CO5	3
	Or				
V (a)	Write c program to read the details of employees working in a company using structures. The details include name, age, date of joining and salary.	5	L3	CO4	3
(b)	What is a function? Differentiate it with recursion .illustrate with an example	5	L4	CO5	3,5